MIL-A-12391B(EA)
20 October 1977
SUPERSEDING
MIL-A-12391A(EA)
31 October 1967

MILITARY SPECIFICATION

APPARATUS, FILLING, FIELD,

LAND MINE, M2

This specificaiton is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of filling apparatus for land mine3s.

2. APPLICABLE DOCUMENTS

2.1 Governmetn documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specificaiton to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636 - boxes, Shipping, Fiberboard.
PPP-C-843 Cushioning Material, Cellulosic.
PPP-P-291 - Paperboard, Wrapping and Cushioning.
PPP-T-60 - Tape, Pressure-Sensitive Adhesive, Waterproof, For Packaging.

FSC 1040

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addresse to : Commander, US Army Armament Research and Development Command, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardizaiton Document Improvement Proposal (SS Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-B-121 - Barrier Material, Greaseprofed, Waterproofed, Flexible.

STANDARDS

FEDERAL

FED-STD-356 - Commercial Packaging of Supplies and Equipment.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

U.S. ARMY ADMENDMENT RESEARCH AND DEVELOPMENT COMMAND

EDGEWOOD ARSENAL

DL21-49-2 - Apparatus, Filling, Field, Land Mine, M2.

(Copies of specifications, standards, drawings and publications required by cotractors-in connection with specific procurement functions should be obtained from the procuring actitity or as directed by the contracting officer.)

3. Requirements

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing D21-49-2 and on the subsidiary drawings shall conform to the specification thereon or specific characteristics set forth on the drawings.

3.1.2 Components. All components shall conform to the specifications and drawings listed on Drawing D21-49-2 and on the subsidiary drawings.

3.2 Assembly. The apparatus shall be assembled as shown on Drawing D21-49-2.

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3.3 Leakage. The pipette (Drawing D21-49-7) shall not leak when tested in accordance with 4.3.1.

3.4 Functioning. The filling apparatus assembly shall be functionally tested to assure that all valves, operating linkage and moveable parts operate as intended when tested in accordance with 4.3.2.

3.5 Workmanship. The apparatus shall be free from contamination (foreign matter) and damage such as cracks or deformation which could effect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all sinspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification shere such inspections are deemed necessary top assure supplies and services conform to prescribed requirements.

4.1.2 Objectie evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the apparatus produced by one manufacturer, at one plant, from the same materials and undr essentially the same manufacturing conditions.

4.2.2 Sampling for examination and nondestructive test. Sampling shall be conducted in accordance with MILSTD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination and nondestructive test. Sample aparatus and the level A preparation for delivery shall be examined in accordance with the classificaiton of defects and with MIL-STD-105.

4.2.3.2 Critical defects. Each apparatus shall be tested for its critical characteristics listed in the classificaiton of defects.

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4.2.3.3 Classification of defects.

(a) Apparatus, Filling, Field, Land Mine M2 (Dwg D21-49-2).

Categories	Defects	Acceptance standards
Critical :		
1	Leakage	4.3.1
2	Functioning of valves, operating linkage and moveable parts	4.3.2
Major:	AQL 1.0 percent defective	
101	Component missing or incorrectly assembled	
102	Workmanship (3.5)	
Minor:	AQL 2.5 percent defective .	
201.	Protective finish missing,	

4.3 Tests.

4.3.1 Leakage. Plug all openings to the pipette, except the opening where the air is to be introduced. The pipette shall be connected to an air pressure source and subjected to an internal pressure of 5 to 1 pored per square inch guage (PSIG) [O. 3515 ±.0703 kilograMs per square centimeter (kg/cm^2)] of air for a minimum period of three minutes. A suitable wetting agent shall be applied to all joints after the pressure Is stabilized. Examine for leakage. Bubble shall indicate leakage. Evidence of leakage shall be cause for rejection.

4.3.2 Functioning. Fill a container with water capable of delivering a minimum of one gallon of water to the receiving Vessel for each test. The container shall be connected to the upper plug valve of the pipette adapter. Place the receiving vessel on the can seat plate to catch the water. Raise the operating handle to the up position. this closes the lower plug valve and opens the upper plug valve. When the air stops escaping from the vent hood, the pipette is filled. Lower the operating handle to the down position. This closes the upper plug valve and opens the lower plug valve. Deliver approximately one gallon of water to the receiving vessel and raise the handle to the up positions. All valves, linkage and moveable parts shall function. Failure of the valves, linkage or moveable parts to function shall be cause for rejection. Pass through the apparatus to dry the assembly upon completion of the test.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging, levels A, B, and commercial (see 6.2).

5.1.1 Level A. All packaging shall be of minimum cube required to contain the items.

5.1.1.1 Cleaning and drying. All metal surfaces of the apparatus and component parts thereof, shall be thoroughly cleaned in accordance with process C-1 of MIL-P-116 and dried in accordance with any applicable drying process of the same specification.

5.1.1.2 Pedstal frame (Dwg D21-49-30). The frame shall be preserved in accordance with method I of MIL-P-116. Preservative oil conforming to P-7 of the same specificaiton shall be applied to all bare metal surfaces of the guide rod and lever fulcrum. The preserved area shall be wrapped wiht barrier material conforming to type I, Grade C, Class 1 or 2 of MIL-B-121.

5.1.1.3 Component parts package (including parts of the pedestal assembly). The component parts consisting of the items listed below shall be preseved and packaged together in accordance with method I of MIL-P-116. Corrosion preventive oil conforming to P-7 of the same specifiecation shall be applied to each part as required. The preserved area of each part shall themn be wrapped with barrier material conforming to type I, Grade A, Class 2 of MIL-B-121. Each item shall be overwrapped with paperoard cushioning material conforming to tpe III, style I of PPP-P-291. The paperboard wrap shall be secured with tape conforming oty class 1 or 2 of PPP-T-60. The component parts shall then be placed within a fiberboard box eonforming to type CF, Class weather resistant, variety SW, Grade V3c, style RSC of PPP-B-636. Cushioning material conforming to typw II, Class C, of PPP-VC-843 shall be used as required to prevent movement of the parts within the container. The box shall be closed in accordance with the appendix of PPP-B-636 for weather resistant boxes.

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Item	Qty	Dwg
Pipette adapter (55 gal drum)	1	B21-49-44
Pipette adapter (1-ton container)	1	B6-19-1
*Plate stopbar	2	A21-49-36
*Can seat plate assembly	1	B21-49-33
Can seat rod	1	A21-49-27
		B21-49-13
Braces (3) (pipet supports),		B21-49-l4
clamps, and brace connectors		B21-49-15
		B21-49-16
Coupling (3/8-in. pipe)	1	B21-49-3
Pipette vent (3/8-in. by 18-1/2 in.)	2	D21-49-3
Drip cup assembly	1	B21-49-25
Valve=43perating handle		
Assembly (two levers)		B21-49-17
Assembly to the operating handle assembly		B21-49-19
Hood vent assembly	1	B21-49-10
*Lever assembly	1	B21-49-35
*Lever stop		B21-49-37
Operating instructions (1) package method IC-3 of MIL-P-116		

*Note. The two plate-stop bars, the can seat assembly, the lever assembly, arid the lever stop, which are packaged with other conponents are part of the pedestal assembly.

5.1.1.4 Accessory set package. The acessory set package consisting of the following items shall be preserved and packaged in the same manner specified in 5.1.1.3.

Item	Quantity
Angle valve adapters for l-ton containers	2
3/4 inch pipe coupling	1
3/4 inch 90° elbow	1
24 lubricant sticks, No. 421, size B (NSN 9150-00-261-8338)	1 box
24 lubricant sticks, no. 542, size B (NSN 9150-00-261-8339)	1 box
3/4 x 1 3/8 inch pipe nipple	1
3/4 x 2 1/2 inch pipe nipple	1
3/4 x 9 inch pipe nipple	2
3/4 x 18 inch pipes	2
3/4 inch gate valve	1
White paste piGMEnt (white lead in oil)	1 pound
1/8 inch hex wrench	1
10 inch pipe wrench	2
18 inch pipe wrench	1
8 inch pipe wrench	1

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5.1.1.5 Pipette body(Dwg D21-49-3). the pipette body with upper and lower plug valves (Dwg B22-62-21), a 3/4-in. by 1 3/8-in. pipe nipple, and the calibrating and filling nozzles (Dwg A21-49-23 and Dwg B21-49-9) assembled to it shall be preserved and packaged in accordance with method 1-A-14 of MIL-P-116. Cushioning material conforming to type 11, Class C of PPP-C-843 shallbe used to assure fittingpackage. *Tape as* specified in 5.1.1.3 shall be used to seal all openings of the pipette body.

5.1.2 Level B. The apparatus shall be preserved and packaged as specified in 5.1.1 except the fiberboard boxes specified in 5.1.1.3 and 5.1.l.4 shall conform to Class domestic, variety SW, Grade 275, style RSC, of PPP_B_636. The boxes shall be closed in accordance with the appendix of PPP-B-636 *for domestic grade boxes. The* pipette body specified in 5 .1.1. shall be preserved and packaged in accordance with method IC-4 of MIL-P-116.

5.1.3 Commercial. The apparatus shall be preserved and packaged in accordance with FED-STD-356.

5.2 Packing.

5.2.1 Level A. The pedestal, component parts, accessory set and pipette body preserved packaged as specified in 5.1.1.2 thru 5.1.1.5 shall be positioned and packed in accordance with Drawings D21-49-55 and B21-49-56.

5.2.2 Level B. The apparatus shall be positioned and packed in the same *manner* Specified *in* 5.2.1, except that the box shall be class 1 of PPP-B-621. All other requirements specified on Drawings D21-49-55 and B21-49-56 remain unchanged.

5.2.3 Commercial. The apparatus shall be packed in accordance with FED-STD-356.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129 and FED-STD-356 as applicable.

6. NOTES

6.1 Intended use. This apparatus is intended for field filling chemical land mines.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date *of* this specification.
(b) Levels of packaging and packing.

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Preparing activity:

Army - EA

Project No. 1040-A115

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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NOTICE OF INACTIVATION
FOR NEW DESIGN

INCH-POUND

MIL-A-12391B(EA)
NOTICE 1
2 September 1995

MILITARY SPECIFICATION

APPARATUS, FILLING, FIELD, LAND MINE, M2

This notice should be filed in front of MIL-A- 12391B(EA)
dated 20 October 1977.

MIL-A- 12391 B(EA) is inactive for new design and is no longer used except for replacement purposes.

Preparing activity:

Army - EA

Project No. 1040-A220

AMSC N/A

FSC 1040

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